

QGIS Skills Review Worksheet

Chapter 01 – Understanding Vector and Raster Data

Q1: Explain the difference between vector and raster data.

Answer:

Vector data represents features using points, lines, and polygons (for example, roads, buildings, and boundaries).

Raster data represents information as a grid of pixels, each with a value (for example, elevation, temperature, or satellite images).

Vector and raster



Vector

- Features with discrete shapes and boundaries (e.g., street, land ownership parcel, well)
- Database management
- Database query and reporting
- Network analysis
- High quality maps



Raster

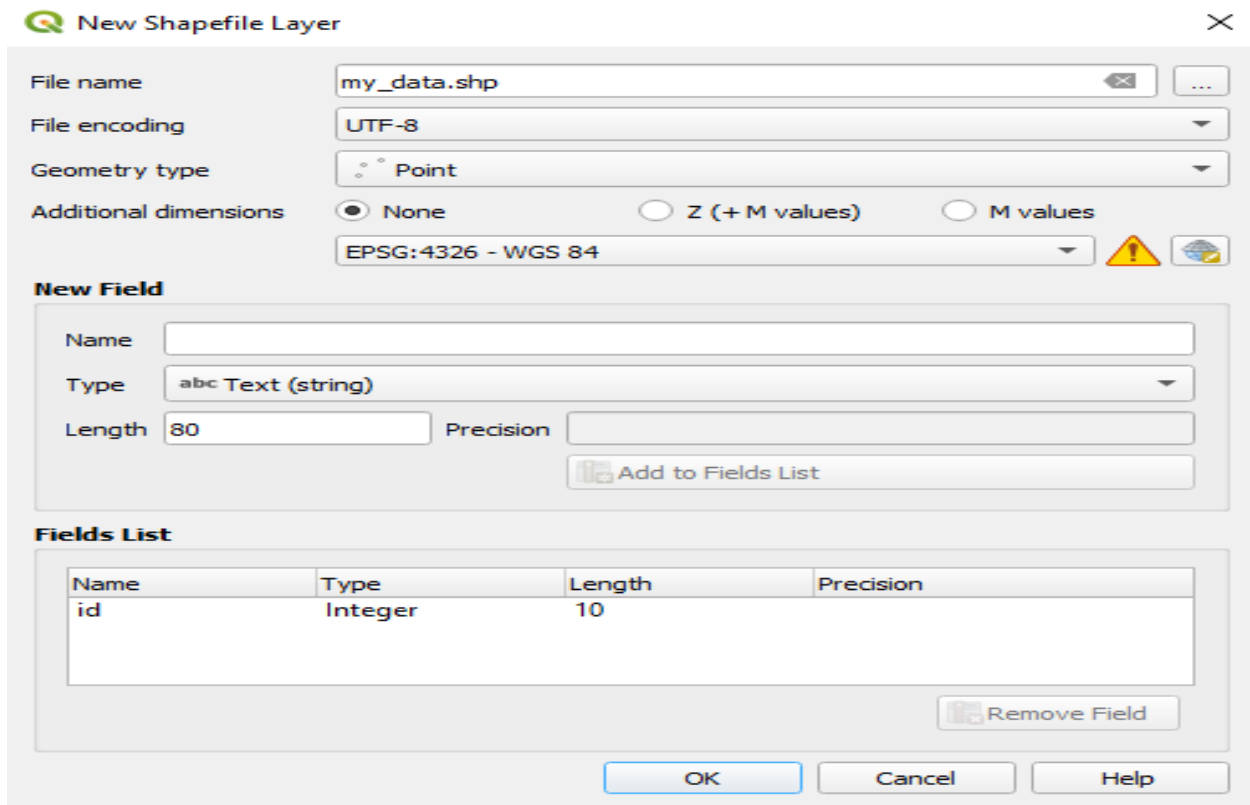
- Continuous surfaces with fuzzy boundaries or with qualities that change gradually over space (e.g., soil, land cover, vegetation, pollution)
- Spatial analysis and modeling (e.g., agricultural suitability)

Chapter 02 – Creating New Layers

Q2: Describe how to define geometry and attributes for a new vector layer.

Answer:

1. Go to **Layer → Create Layer → New Shapefile Layer**.
2. Choose the **Geometry Type** (Point, Line, or Polygon).
3. Add **Attribute Fields** such as Name, ID, or Type.
4. Save the file as .shp.



New Shapefile Layer

File name: my_data.shp

File encoding: UTF-8

Geometry type: Point

Additional dimensions: ☒ None ☐ Z (+ M values) ☐ M values

CRS: EPSG:4326 - WGS 84

New Field

Name:

Type: abc Text (string)

Length: 80 Precision:

Add to Fields List

Fields List

Name	Type	Length	Precision
id	Integer	10	

Remove Field

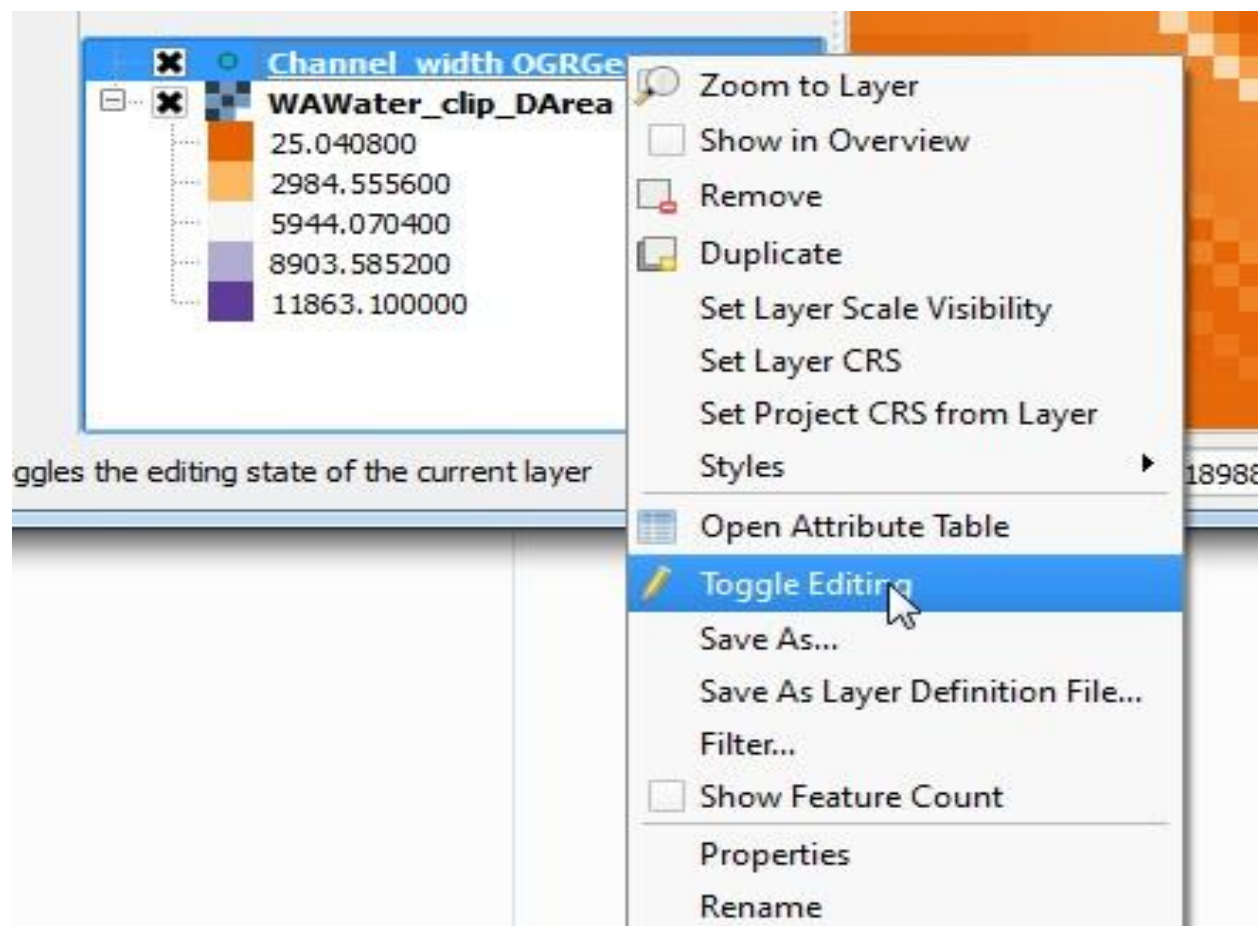
OK Cancel Help

Chapter 03 – Editing and Managing Layers

Q3: How do you edit and save changes to a vector layer?

Answer:

1. Select the layer from the Layers panel.
2. Click the **Toggle Editing** tool (pencil icon).
3. Use editing tools to **add, move, or delete features**.
4. When finished, click **Save Edits** and then **Toggle Editing** again to stop.

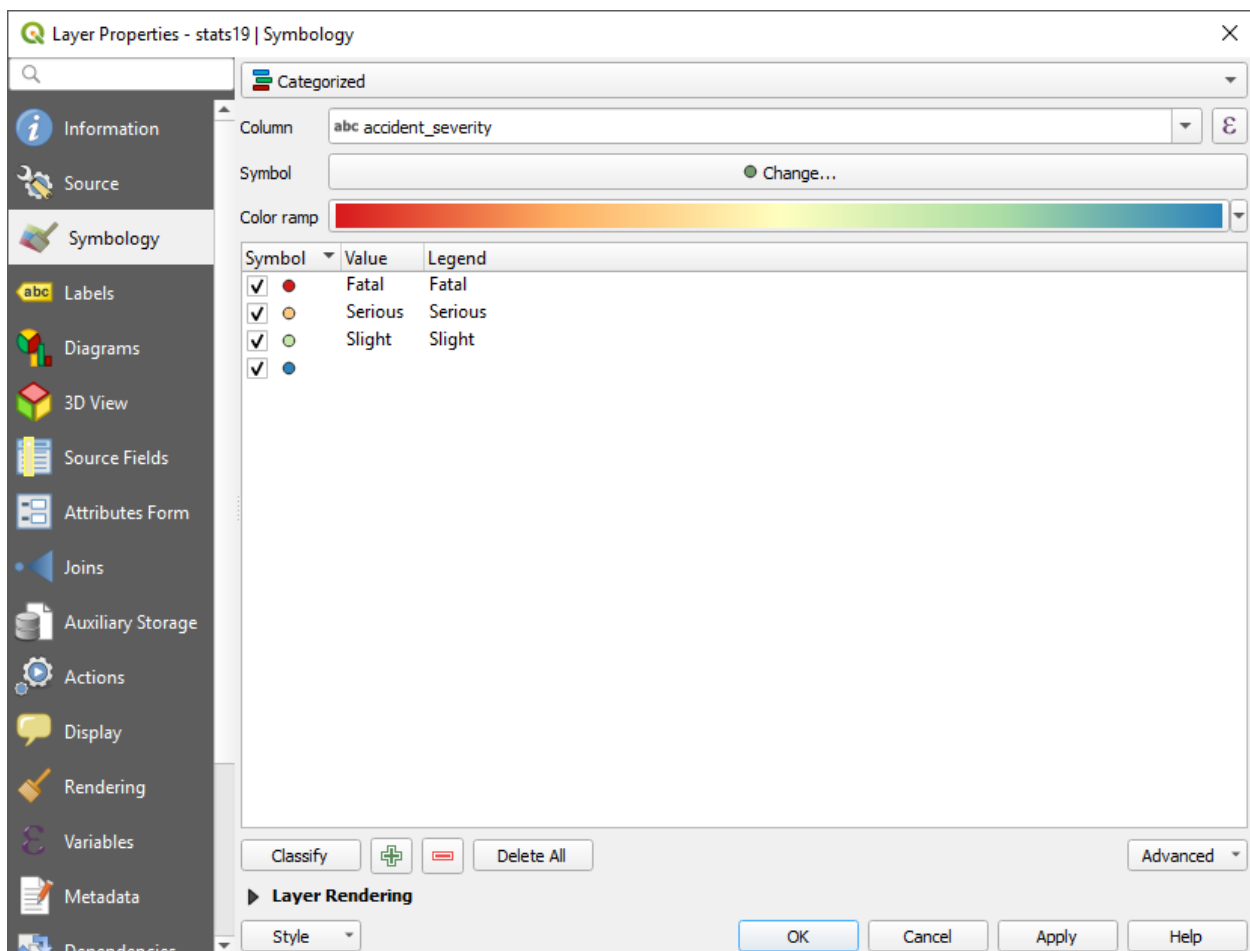


Chapter 04 – Styling and Symbology

Q4: How can you change the symbology (style) of a layer?

Answer:

1. Right-click the layer → **Properties** → **Symbology**.
2. Choose **Single Symbol**, **Categorized**, or **Graduated** style.
3. Select colors, size, or patterns to display data visually.
4. Click **Apply** and **OK**.



Chapter 05 – Working with Attribute Tables

Q5: Explain how to perform an Attribute Join in QGIS.

Answer:

1. Right-click the target layer → **Properties** → **Joins** tab.
2. Click the **+** button to add a join.
3. Choose the **Join Layer** and match the **Join Field** with the **Target Field**.
4. Click **OK** and open the Attribute Table to see the merged data.

Add Vector Join

Join layer: trees

Join field: 1.2 cat

Target field: 1.2 cat

☒ Cache join layer in virtual memory

☐ Create attribute index on join field

☐ Dynamic form

☐ Editable join layer

☐ Upsert on edit

☐ Delete cascade

☒ Joined Fields

cat

☒ VEGDESC

☒ VEG_ID

☐ F_CODEDESC

☐ F_CODE

☒ AREA_KM2

☐ Custom Field Name Prefix

trees_

LinkedIn Post

https://www.linkedin.com/feed/update/urn:li:activity:7381408189608718336?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEVjVk0BqHVMXt8wz8gUvE0Evt_AZzhhx58